Safety and Lab Conduct Guide

Keeping outreach experiments safe, clean and responsible

THE GOLDEN RULES

- **Think first.** Every activity has been planned to be safe *if the steps are followed*. Do not improvise with chemicals or microbes.
- **Only approved materials.** Use sealed, premade plates and harmless lab strains. Never culture samples from the nose, mouth, toilets, wounds or animals.
- **Never taste, eat or drink** in the work area. Do not put anything from the experiment near the mouth.
- **Tell the teacher at once** about any spill, cut, splash or broken glass.

1. Before the session

- Read the activity from start to end and do a **trial run** yourself.
- Fill in a short **risk assessment** (template below).
- Get **permission** from the school and consent from parents or guardians, including for photographs.
- Check that a **first-aid kit**, a sink or water, and a fire blanket or extinguisher are close by.
- Make sure there are enough adults: at least one supervisor for each group of about 10 students.

2. Personal protection and behaviour

- **Wear** lab coats or old clothes, closed shoes, **gloves** and **goggles** where the activity says so. Tie back long hair and secure loose clothing.
- Wash hands with soap and water **before and after** every practical.
- Keep bags and food away from the benches. Cover cuts with waterproof plasters.
- Walk, do not run. Keep the work area clear and quiet.

3. Working with microbes

Our activities use only Biosafety Level 1 (BSL-1) organisms: harmless lab strains such as E. coli DH5α, and baker's yeast. Even so, treat every culture as if it could be harmful.

Good sterile technique

- Disinfect the bench with 70 % ethanol or a disinfectant before and after work. Let it dry — and keep ethanol **away from flames**.
- Keep plates closed as much as possible; open them only as far as needed, for as short a time as needed.
- Seal plates with tape after inoculation. **Never open** a plate after incubation.
- Incubate plates **upside down**.

- Do not touch your face, phone or hair during practical work.

Disposal of cultures

- **Autoclave** all plates, tubes and tips at 121 °C (15 psi) for at least 15–20 minutes, **or**
- flood with fresh **10 % household bleach** (1 part bleach to 9 parts water) and leave for at least **30 minutes**; then bag and discard.
- Never pour cultures, antibiotic plates or antibiotics down the sink: they can spread resistance.
- Disinfect benches, pipettes and loops. Wash hands.

If something spills

1. Tell the teacher; keep others away.
2. Wearing gloves, cover the spill with paper towels.
3. Pour disinfectant (10 % bleach) over the towels. Wait 20–30 minutes.
4. Pick up the towels with tongs or gloved hands and put them in a biohazard bag.
5. Wipe the area again with disinfectant and wash hands.

4. Activity-by-activity checklist

Activity	Main hazards	What to do
Agar plate art / streaking	Unknown skin bacteria; fire from loop sterilisation; ethanol	Sealed plates; room-temperature incubation for hand prints; never open plates; tie back hair; ethanol away from flame; disinfect before disposal
Yeast balloons	Spills, hot water; allergies	Warm (not hot) water; wipe up spills; no tasting; check allergies
Fingerprints: graphite and ink	Staining; dust	Gloves; aprons; wipe surfaces
Fingerprints: iodine fuming	Iodine vapour is toxic to breathe and irritating; stains	Teacher only. Fume hood or open-air; gloves and goggles; students stay back; keep lid closed; do not heat directly
Catalase / hydrogen peroxide	3 % peroxide irritates eyes and skin	Goggles; wash with water if splashed; do not drink
Biomagnification sweets	Food allergies; hygiene	Do not eat sweets handled in the activity; check allergies (nuts, colourants)
Gel electrophoresis (if demonstrated)	Electric shock; hot agarose; DNA stains	Teacher only; keep water away from power; use safer DNA stains; gloves; hot agarose in heat-proof gloves
Plasmid cloning (demonstration or theory)	Antibiotic resistance; genetically modified bacteria	For school level this is discussion or demonstration only, by trained staff in a proper lab; never release bacteria into the environment

5. Responsible science

- **Respect for living things:** use only what you need; do not harm animals.

- **Biosafety:** engineered microbes must stay in the lab, and must be inactivated before disposal. Teams plan *containment* — for example a "kill switch" that makes cells die outside the intended environment.
- **Honesty:** record what really happened, even if the result is "wrong". Unexpected results are often the most interesting ones.
- **Respect for people:** treat everyone fairly; share equipment and roles; use inclusive language; ask permission before taking or sharing photographs.

6. First aid basics

Problem	First response
Chemical in eye	Rinse immediately with clean running water for at least 10–15 minutes; get medical help.
Chemical on skin	Remove contaminated clothing; rinse with plenty of water.
Small cut	Wash with soap and water; cover with a plaster; tell the teacher.
Burn (flame or hot water)	Cool under cool running water for at least 10 minutes; do not apply ointment; get help for larger burns.
Swallowed chemical or culture	Do not induce vomiting; rinse the mouth with water; call a doctor or poison centre and bring the label.
Inhaled vapour	Move to fresh air at once; get medical help if there is coughing or breathing trouble.

EMERGENCY NUMBERS
Write your local emergency numbers and the nearest hospital on the wall of the room where the activity takes place.

7. Risk assessment template

Question	Your answer
Activity and date	
Number of students and adults	
What could go wrong?	
Who could be harmed?	
What will we do to prevent it?	
What will we do if it happens?	
Approved by	